ExternalStream#

https://pytorch.org/docs/stable/generated/torch.cuda.ExternalStream.html

Description:

Wrapper around an externally allocated CUDA stream. This class is used to wrap streams allocated in other libraries in order to facilitate data exchange and multi-library interactions. This class doesn’t manage the stream life-cycle, it is the user responsibility to keep the referenced stream alive while this class is being used. stream_ptr (int) – Integer representation of the cudaStream_t value. allocated externally.

Function Signature

```
class torch.cuda.ExternalStream(stream_ptr, device=None,
**kwargs)[source]#

Parameters

query()[source]#

Returns

Return type
```

Returns:

Recorded event.

Notes & Warnings

NOTE Note This class doesn't manage the stream life-cycle, it is the user responsibility to keep the referenced stream alive while this class is being used.

NOTE Note This is a wrapper around cudaStreamSynchronize(): see CUDA Stream documentation for more info.

NOTE Note This is a wrapper around cudaStreamWaitEvent(): see CUDA Stream documentation for more info. This function returns without waiting for event: only future operations are affected.

NOTE Note This function returns without waiting for currently enqueued kernels in stream: only future operations are affected.

Detailed Documentation

Documentation

Wrapper around an externally allocated CUDA stream.

This class is used to wrap streams allocated in other libraries in order to facilitate data exchange and multi-library interactions.

This class doesn't manage the stream life-cycle, it is the user responsibility to keep the referenced stream alive while this class is being used.

stream_ptr (int) – Integer representation of the cudaStream_t value. allocated externally.

device (torch.device or int, optional) – the device where the stream was originally allocated. If device is specified incorrectly, subsequent launches using this stream may fail.

Check if all the work submitted has been completed.

A boolean indicating if all kernels in this stream are completed.

event (torch.cuda.Event, optional) – event to record. If not given, a new one will be allocated.

Wait for all the kernels in this stream to complete.

This is a wrapper around cudaStreamSynchronize(): see CUDA Stream documentation for more info.

Make all future work submitted to the stream wait for an event.

event (torch.cuda.Event) – an event to wait for.

This is a wrapper around cudaStreamWaitEvent(): see CUDA Stream documentation for more info.

This function returns without waiting for event: only future operations are affected.

Synchronize with another stream.

All future work submitted to this stream will wait until all kernels submitted to a given stream at the time of call complete.

stream (Stream) – a stream to synchronize.

This function returns without waiting for currently enqueued kernels in stream: only future operations are affected.